THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 7: **Coordinate Geometry** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The distance of the point $(5, -12)$ from the origin is: 1

(a) 7 **(b)** 13 **(c)** 17 **(d)** 12

2. The distance between $(0, 0)$ and $(36, 15)$ is: 1

(a) 39 **(b)** 21 **(c)** 51 **(d)** 40

3. The mid-point of the segment joining $(-2, -2)$ and $(2, -4)$ is: 1

(a) $(0, -3)$ **(b)** $(0, 3)$ **(c)** $(-3, 0)$ **(d)** $(2, -3)$

4. The point $(-3, 0)$ lies: 1

(a) on the y-axis **(b)** on the negative x-axis

(c) in Quadrant II **(d)** in Quadrant III

5. A student states: “The mid-point of a segment divides it in the ratio 2 : 1.” The correct response is: 1

(a) Correct

(b) Incorrect — the mid-point divides it in the ratio 1 : 1

(c) Correct only on the x-axis

(d) Cannot be determined

6. The point $P(2, 4)$ divides the join of A(0, 2) and B(3, 5) in the ratio: 1

(a) 2 : 1 **(b)** 1 : 2 **(c)** 1 : 1 **(d)** 3 : 2

7. For $P(-1, 1)$ and $Q(3, 4)$, Student A gets $PQ = 5$ and Student B gets $PQ = \sqrt{7}$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

8. If the mid-point of the segment joining $(x, 4)$ and $(6, y)$ is $(4, 5)$, then (x, y) is: 1

(a) $(2, 6)$ **(b)** $(6, 2)$ **(c)** $(2, 5)$ **(d)** $(4, 6)$

9. The fourth vertex D of parallelogram ABCD with A(1, 2), B(3, 5), C(7, 7) is: 1

(a) $(5, 4)$ **(b)** $(4, 5)$ **(c)** $(9, 10)$ **(d)** $(3, 4)$

10. The points $(0, 0)$, $(3, 0)$ and $(0, 4)$ form a triangle with perimeter: 1

(a) 12 **(b)** 7 **(c)** 10 **(d)** 14

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the value of x if the distance between $(x, -1)$ and $(3, 2)$ is 5. 2

12. Find the coordinates of the point dividing the join of $(-1, 7)$ and $(4, -3)$ in the ratio 3 : 2. 2

13. Find the point on the x-axis which is equidistant from $(2, -5)$ and $(-2, 9)$. 2

14. A student claims that the points A(1, 5), B(2, 3) and C(−2, −11) are collinear. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the ratio in which the point $P\left(\frac{3}{4}, \frac{5}{12}\right)$ divides the segment joining $A\left(\frac{1}{2}, \frac{3}{2}\right)$ and $B(2, -5)$. 3

16. If the point (x, y) is equidistant from $(7, 1)$ and $(3, 5)$, show that $x - y = 2$. 3

17. Find the type of quadrilateral formed by A(−1, −2), B(1, 0), C(−1, 2) and D(−3, 0). Give reasons. 3

OR

Show that A(4, 7), B(4, −1) and C(0, 3) form an isosceles triangle. Is it also right-angled? Find its perimeter. 3

18. If $P(2, p)$ is the mid-point of the segment joining $A(6, -5)$ and $B(-2, 11)$, find p. Also find the length AB. 3

OR

Find the coordinates of the points of trisection of the segment joining A(2, −2) and B(−7, 4). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the coordinates of the point which divides the segment joining A(2, −3) and B(12, 7) in the ratio 2 : 3. If the x-coordinate of A is increased by 4 and the y-coordinate of B is decreased by 5, find the coordinates of the point dividing the new segment in the same ratio. 5

OR

Find the ratio in which the line segment joining A(1, −5) and B(−4, 5) is divided by the x-axis. Also find the point of division, and verify using the section formula for the x-coordinate. 5

20. **Case study:** In a school ground ABCD, lines are drawn 1 m apart along AD. 100 flower pots are placed 1 m apart along AD. Niharika runs $\frac{1}{4}$ of AD on the 2nd line and posts a green flag; Preet runs $\frac{1}{5}$ of AD on the 8th line and posts a red flag. 5

(i) Write the positions of the flags.

(ii) Find the distance between the flags.

(iii) Rashmi posts a blue flag exactly halfway between them. Where should she post it?

OR

The vertices of $\triangle ABC$ are A(4, 6), B(1, 5) and C(7, 2). 5

(i) Find the mid-points D of AB and E of AC.

(ii) Show that $DE = \frac{1}{2}BC$.

(iii) What does this verify?

END OF QUESTION PAPER — **Total: 40 Marks**